

# IKECHUKWU JOHN CHUKWU

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## EDUCATION

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- MSc. in Computer Science (Erasmus Mundus)** Sept. 2023 – Sept. 2025
- Ss. Cyril and Methodius University, Skopje, North Macedonia
- MSc. in Cyber Security (Erasmus Mundus)** Sept. 2023 – Sept. 2025
- Kadir Has University, Istanbul, Turkiye
- Bachelor of Technology in Mathematics**
- Advisor: Prof. M. C Obi. Sept. 2013 - Sept. 2018
- Federal University of Technology, Nigeria

## RESEARCH EXPERIENCES

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### \_Conference Proceedings

- (1) **Ikechukwu, J. C.**, et al., *Integrating Cryptographic Techniques to Protect AI Systems and Data*. In: *Proceedings of the 16th ICT Innovations Conference*, pp. 54–70, 2024.  
*Contribution:* Led analysis of cryptographic mechanisms for securing AI pipelines and evaluated implementation trade-offs.
- (2) Jose Luis Cano Garcia, Izuchukwu Patrick Udechukwu, Isiaq Bolaji Ibrahim, **Ikechukwu, J. C.**, Hasan Dağ, Vesna Dimitrova, & Elissa Mollakuqe, *Securing AI Systems: A Comprehensive Overview of Cryptographic Techniques for Enhanced Confidentiality and Integrity*. In: *13<sup>th</sup> Mediterranean Conference on Embedded Computing (MECO)*, Budva, Montenegro, pp. 1–8, 2024.  
*Contribution:* Conducted literature review and synthesis of the cryptographic approaches for AI systems.

### \_Journal Articles

- (1) Mollakuqe E., Rexhepi S., Bunjaku R., **Ikechukwu J. C.**, *Algorithm for Key Transparency with Transparent Logs*. Open Research Europe, 4:163, 2024.  
*Contribution:* Conducted literature review.
- (2) Imamverdiev, Y., Bagirov, E., & **Ikechukwu, J. C.** *Detecting Obfuscated Malware on Windows Using Ensemble Learning Methods*. Computer Science and Automation, 24(1), 99–124, 2025.  
*Contribution:* Conducted literature review.

### \_Thesis

- (1) *Assessing Lightweight Ciphers: An Evaluation Framework* (Master's Thesis)  
*Contribution:* Designed and implemented a modular benchmarking framework in C to evaluate lightweight cryptographic ciphers, with emphasis on nonce misuse and resistance properties.
- (2) *Some Applications of Number Theory to Public-Key Cryptography* (Undergraduate Thesis)  
*Contribution:* Studied and wrote about number-theoretic foundations of public-key cryptosystems, including modular arithmetic and their applications in encryption schemes.

### \_Report(s)

- (1) The European Approach for Quality Assurance of Joint Programs Report.  
*Contribution:* Co-authored sections on Student Support Services and Transparency of Documentation, and participated in the evaluation of cybersecurity education programs under EHEA standards.

### \_Talks & Presentations

- (1) Delivered a lecture on “**Two-Factor Authentication, Cookies, and SSO**” to graduate students at the Faculty of Computer Science (UKIM). May 2025  
*Contribution:* Presented authentication concepts, usability, and security in real-world systems.  
Lecture link
- (2) Delivered a presentation on blockchain systems and another on privacy-preserving technologies for healthcare data privacy as a research fellow at MuseMatrix.

## HONORS AND AWARDS

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- Erasmus Mundus Joint Master's Scholarship in Applied Cybersecurity (CyberMACS) 2023 - 2025
- ORSHIN Summer School on Open-source ReSilient Hardware and software for Internet of thiNgs, Greece Sept. 2025

## EMPLOYMENT HISTORY

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- **Cyber Security Engineer** *Dec. 2024 – Jul 2025. March 2026 – Present*  
*Gigatt, Skopje, North Macedonia*
  - Ensuring secure web development and delivering Cybersecurity awareness lectures and updates to staff.
- **Quality Assurance Evaluation Expert** *Sept. – Nov. 2025*  
*HCERES, France*
  - Served as an external evaluator for the Cyberus joint program under the European Approach for Quality Assurance of Joint Programs.
  - Contributed two sections: Student Support Services and Transparency of Documentation, to the published evaluation report.
- **Quality Assurance Student Representative** *Sept. 2023 - 2025*  
*CyberMACS.eu, Turkiye*
  - Developed practical expertise in quality assurance methodologies, focusing on cybersecurity education across consortium institutions of the CyberMACS EU project. Certificate of Recognition
- **Cyber Security Consultant** *Jul. – Oct. 2024*  
*KPMG, Istanbul, Turkiye*
  - Served as a consultant for a bank in Amman, Jordan, by assessing IT risks, analyzing data, and developing security strategies based on the NIST framework. Supervisor: Betül Uzun
- **Technical Instructor** [Github Repo] | Link *Aug. 2022 – 2023*  
Hands-on Institute of Information Technology (HIIT), Lagos, Nigeria (hiitplc.com)
  - Taught HTML, CSS, JavaScript, Git, and responsive design principles.
  - Supported students in debugging assignments and improving problem-solving skills.
- **High-School Mathematics Teacher (12<sup>th</sup> Grade) | Link** *Aug. 2021 – 2022*  
Department of Mathematics, Holy Rosary College, Enugu, Nigeria (hrcenugu.com)
  - Prepared students for the WAEC Mathematics examination.
  - Skills gained: Classroom management, tutoring, grading exercises, and providing comprehensive feedback.

## FURTHER TRAINING

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- **AWS GameDay Participant (Europe) | Amazon Web Services** *Mar. 17, 2026*
  - Utilized AWS services, including VPC, CloudWatch, GuardDuty, and Systems Manager to monitor infrastructure and investigate threats.
  - Collaborated in a team setting to identify misconfigurations and resolve security challenges within **two hours**.
- AWS Cloud Practioner (Certificate). *Mar. 2026*
- Mathematics of Machine Learning Specialization at Coursera.org | Deeplearning.AI (Certificate) *Apr. 2024*
- Applied Data Science Lab | World Quant University (Certificate) *Aug. 2024*
- University Teaching at The University of Hong Kong with Coursera.org (Certificate). *Jun. 2024*

## CURRENT PROJECTS

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1. **Private KV Cache for Autoregressive LLM Inference under FHE** (In Progress) : Implementing a homomorphic encryption-based key-value cache for autoregressive transformer inference using CKKS/OpenFHE. Motivated by open problems in private LLM inference identified in recent literature (Lim et al., EDBT 2026). Github repo

## TECHNICAL TOOLS

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Python, PyTorch, Numpy, Pandas, MongoDB, SQL, Linux/Unix (Bash Scripting), JavaScript/TypeScript, HTML, CSS, L<sup>A</sup>T<sub>E</sub>X, C, Rust, Haskell, Docker, AWS CLI, AWS Cloud Infrastructure